OMAR BOUATTOUR

AI and DATA ENGINEERING STUDENT

Sfax, Tunisia • +216 54 866 800
omar.bouattour.03@gmail.com • linkedin.com/in/bouattour-omar • github.com/OmarBouat

Summary

Final-year Data Engineering student focused on AI Engineering, LLM inference, and AI infrastructure, with hands-on experience in vLLM, Kubernetes, GPU serving, and distributed inference. Seeking a 6-month end-of-studies internship starting February 2027.

Skills

Technical Skills:

Programming Languages: C/C++, Python, Java, JS/TS, SQL/PLSQL

Data Engineering: Data Pipelines, Data Warehousing, Distributed Systems, Hadoop, IoTDB, Spark, Airflow

AI Engineering: Deep Learning, TensorFlow, PyTorch, Scikit-learn, vLLM inference engine, LangChain

Tools & Frameworks: Docker, Kubernetes, Ansible, Power BI, Git, Jira

Soft Skills: Leadership, Mentoring, Cross-Team Collaboration, Communication

Education

Engineer's Degree in Computer Science **09/2024 – Present**
Faculty of Sciences Sfax, University of Sfax, Tunisia

Integrated Preparatory Cycle in Computer Science **09/2022 – 06/2024**
Faculty of Sciences Sfax, University of Sfax, Tunisia

Experiences

AI / LLM Infrastructure Internship – Data Engineering and Semantics Unit, FSS **06/2026 – 08/2026**

- Deployed and operated GPU-based LLM inference services using vLLM and Kubernetes, exposing OpenAI-compatible APIs for local model serving.
- Served and evaluated Qwen and GPT models across different GPU configurations, experimenting with Tensor Parallelism, Pipeline Parallelism, Prefix Caching, and Multi-LoRA.
- Designed performance experiments measuring TTFT, TPOT, end-to-end latency, throughput, concurrency, and KV-cache utilization under different workloads.
- Built an observability stack with Prometheus and Grafana to monitor inference performance, GPU utilization, request behavior, and serving metrics.
- Integrated an LLM Gateway for centralized API access, model routing, and usage management across local AI applications.
- Integrated Open WebUI and local coding agents with the inference infrastructure, enabling developers to consume locally hosted LLMs for code generation, analysis, and debugging.

Summer Internship at Data Engineering and Semantics Unit, FSS **06/2025 – 08/2025**

- Engineered an IoT data platform processing real-time device telemetry using Python, MQTT, Redis, and Apache IoTDB.
- Deployed cloud-based data infrastructure with PostgreSQL, Docker, and Kubernetes for reliable and secure IoT data management.

Summer Internship at ArtiDev **07/2024 – 08/2024**

- Built web applications using ReactJS, .NET, and SQL Server.
- Contributed to Agile sprints, ensuring timely delivery and process improvement.

Projects

LLM Serving Platform | vLLM, LiteLLM, LangChain, Kubernetes, Grafana **04/2026 – 05/2026**

- Built a production-oriented LLM serving platform on Kubernetes using vLLM, exposing Qwen2.5-3B-Instruct through an OpenAI-compatible API.
- Integrated chat, API gateway, SQL agent, multimodal, and monitoring components around a centralized GPU inference backend.
- Developed real-time observability with Prometheus and Grafana to track latency, TTFT, throughput, KV cache usage, and request behavior.
- Evaluated performance under concurrent workloads and compared Prefix Caching, Tensor Parallelism, Pipeline Parallelism, and Multi-LoRA serving.

Kubernetes Platform with End-to-End CI/CD

- Engineered a self-hosted Kubernetes platform with 4 nodes using Vagrant, Ansible, and containerd, enabling reproducible infrastructure provisioning.
- Built an end-to-end CI/CD pipeline using Gitea, image build, Nexus, and Kubernetes to automate application deployment from code commit to cluster rollout.

Vegetable Image Classification | TensorFlow, Keras

- Trained and evaluated MobileNetV2, ResNet50, and a custom CNN for classifying 36 fruit and vegetable categories.
- Achieved 97% accuracy and analyzed trade-offs between accuracy, inference speed, and model size.

Crop Recommendation System | Python, Scikit-learn, Flask

- Built a crop recommendation ML pipeline using soil and climate features, comparing Logistic Regression, Random Forest, SVM, and Gradient Boosting with cross-validation, and deployed a Flask prediction app.

Leadership & Achievements

- **CPC FSS Volunteering & Leadership**
  - **Mentor** **2025–Present**
    Mentor students in algorithms, data structures, and competitive programming.
  - **Lead** **2024–2025**
    Coordinated training sessions, contests, and club initiatives.
  - **HR Manager** **2023–2024**
    Managed onboarding, internal communication, and member coordination.
- **Competitive Programming:** Codeforces Specialist, ranked #60 in Tunisia; 30th place in TCPC 2024; 14th place in EPT Solving Cup 6.0; 3rd place in SolveIT HackJam 2.0; 1st place in Hello World 6.0.
- **Hackathon Awards:** 1st place in Technova 1.0 and Fit-Tech 1.0; 2nd place in MasterFaster 2.0; 3rd place in I2I 1.0 and TRC 2.0.

Certifications

- IBM Data Engineering Foundations Specialization.
- Computer Vision Fundamentals with Google Cloud.

Languages

Arabic: Native | English | French